

Indecomposable polytopes & Rays of the submodular cone

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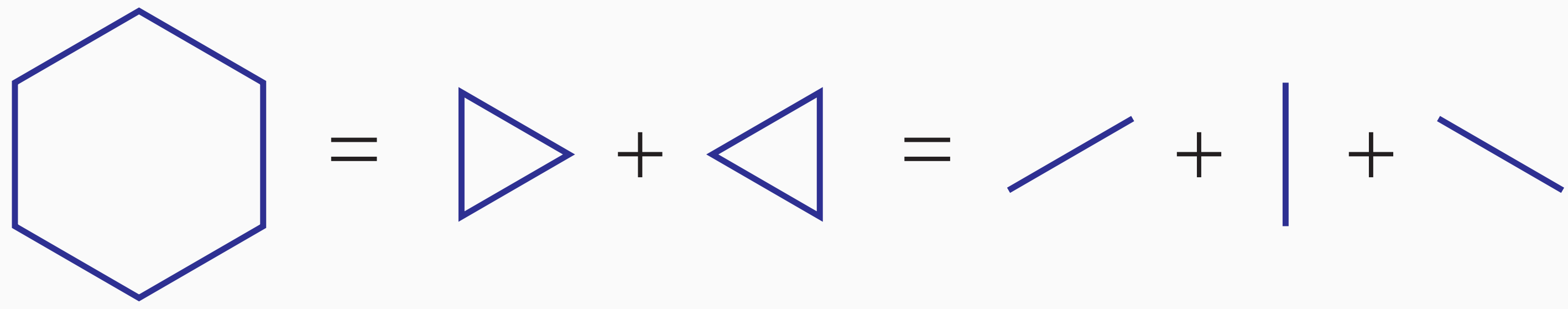


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(In)decomposability of polytopes

Minkowski sum: $Q + R = \{q + r; q \in Q, r \in R\}$

Indecomposable: If $P = Q + R$, then $Q = \lambda P + t$ for some $\lambda > 0$, $t \in \mathbb{R}^d$



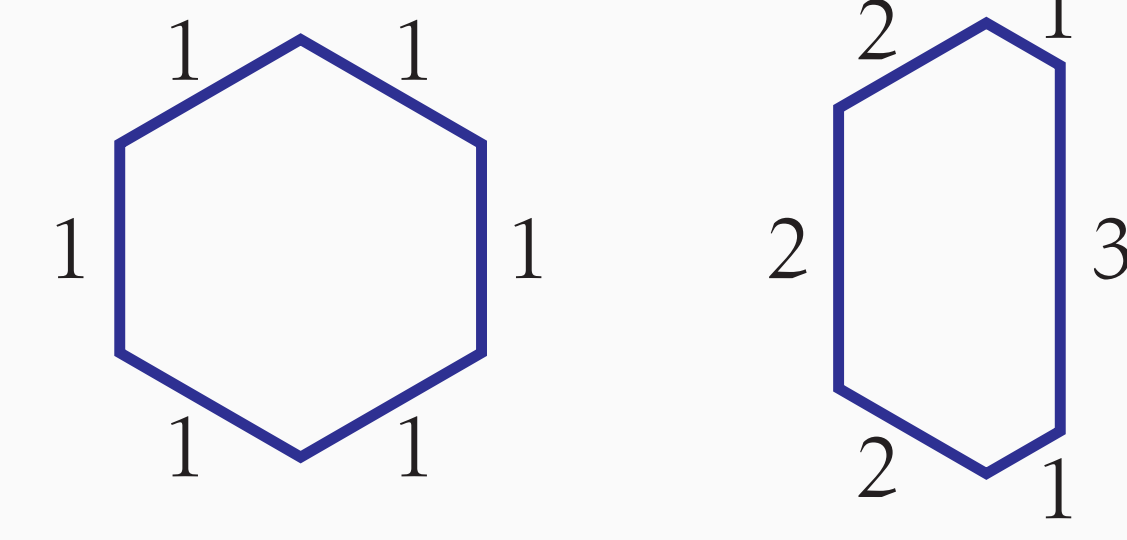
Question A: How to check if P is indecomposable?

Parameterizing deformations of P

Deformations of P : any Q s.t. $\lambda P = Q + R$ for some $\lambda > 0$ and R

$\longleftrightarrow$ edges of Q parallel to edges of P , but not same length

$\longleftrightarrow$ *edge-length vector*: λ_e for each edge e of P , $\lambda_e(Q) = \frac{\text{length}(e \in Q)}{\text{length}(e \in P)}$



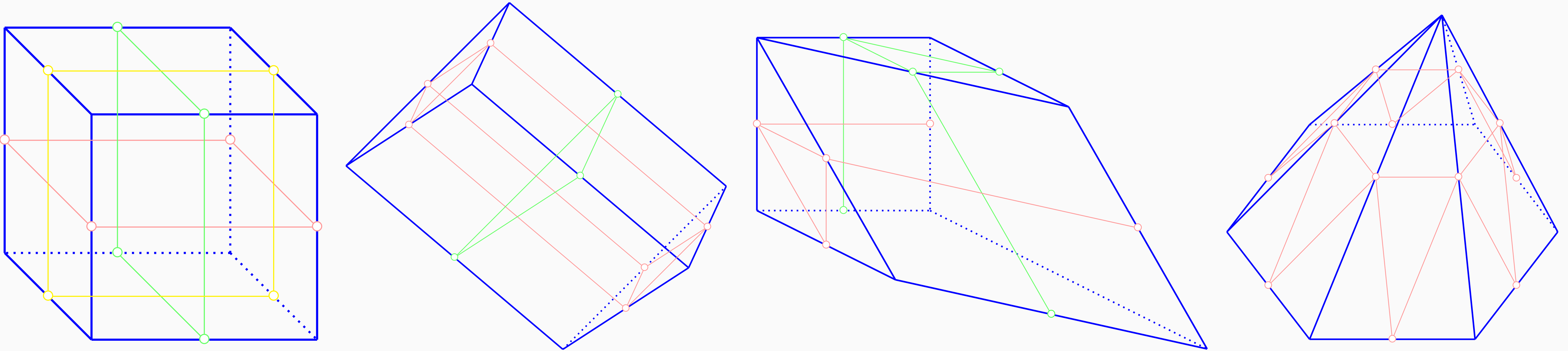
Cycle eqn.: if cycle $e_1, \dots, e_r$, then for all Q : $\lambda_{e_1}(Q)\vec{e}_1 + \dots + \lambda_{e_r}(Q)\vec{e}_r = 0$

Edge-length dependencies graph: new criteria for indecomposability

Dependent edges e, f : for all deformations Q of P , $\lambda_e(Q) = \lambda_f(Q)$

informally: “the length of e can be deduced from the length of f ”

Examples of dependent edges: edges of a triangle; edges of a cycle whose convex hull is a simplex; edges opposite in a parallelogram or a trapezoid...



Graph of edge-length dependencies $ED(P)$: nodes = edges of P ; arcs ef if e and f dependent edges

THM. $ED(P)$ is a clique $\iff ED(P)$ is connected $\iff P$ is indecomposable

$\dim \text{IDC}(P) \leq |cc(ED(P))|$

THM. $\exists$ subset S of dependent edges, forming a connected sub-graph of the graph of P , such that S touches every facet $\implies P$ indecomposable

Application: Re-derive almost all previous indecomposability criteria

Application: Study decomposability products of polytopes $P \times P'$

Edmonds's problem

Deformed permutahedron: all edge directions $e_i - e_j$

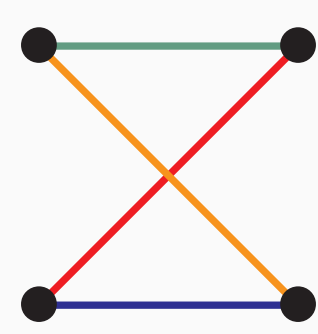
Edmonds '70 problem: Find (all) indecomposable deformed permutahedra!

Nguyen '86: give $2^{n-1/2 \log n + O(1)}$ indecomposable deform. permutahedra via connected matroids

Question B: How to create indecomposable deformed permutahedra, not matroid polytopes?

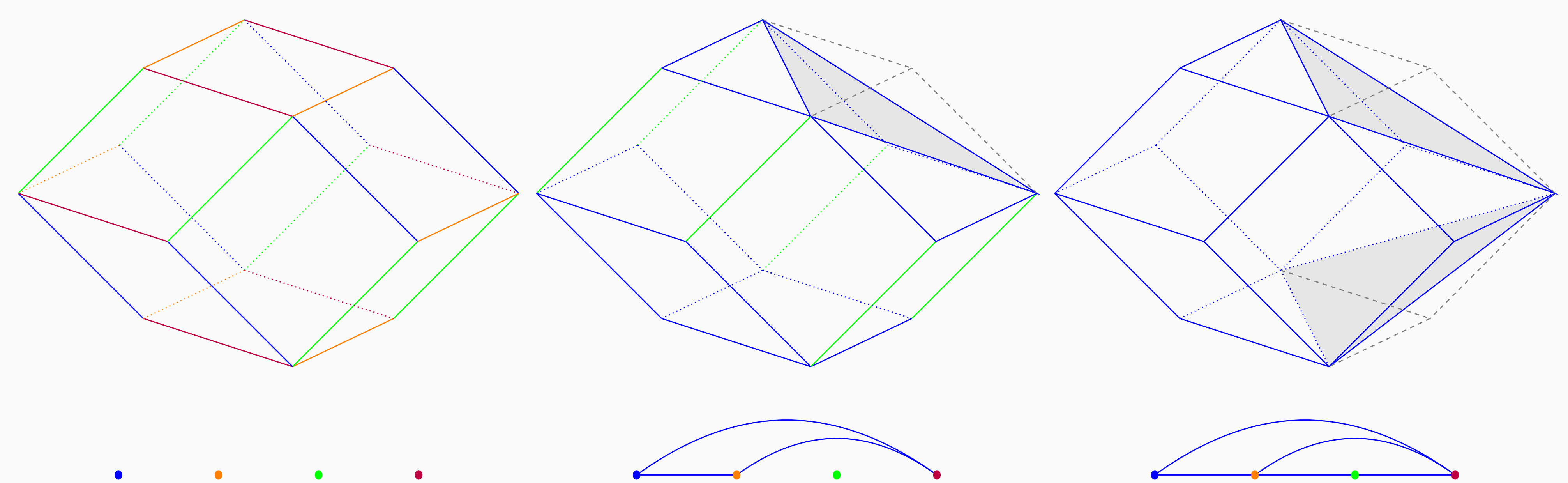
Idea: use *graphical zonotope* of graph G :

$$Z_G := \sum_{i,j \in E} [e_i, e_j] \quad K_{2,2} =$$



Vertices $Z_G \longleftrightarrow$ acyclic orientations of G

Truncating zonotopes



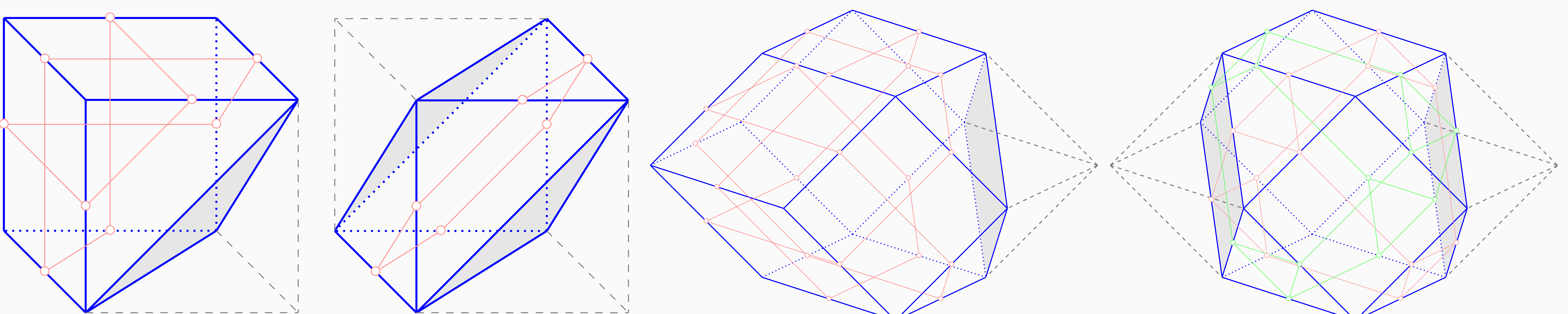
Application: Pick a zonotope Z + remove vertices = Z' $\rightarrow$ easy to tell if Z' is indecomposable
Idem for stacking vertices

New rays of the submodular cone: truncated graphical zonotopes of complete bipartite graphs

Works *only* when $G = K_{n,m}$ complete bipartite graph

$Z_{n,m}^\circ, Z_{n,m}^{\circ\circ}$: graphical zonotope $Z_{K_{n,m}}$, remove 1 or 2 specific vertices

$$Z_{n,m}^\circ := Z_{K_{n,m}} \cap \left\{ x \in \mathbb{R}^{n+m}; \sum_{j=1}^m x_{b_j} \leq nm - 1 \right\} \quad \text{and} \quad Z_{n,m}^{\circ\circ} := Z_{n,m}^\circ \cap \left\{ x \in \mathbb{R}^{n+m}; \sum_{i=1}^n x_{a_i} \leq nm - 1 \right\}$$



THM. $Z_{n,m}^\circ$ and $Z_{n,m}^{\circ\circ}$ are indecomposable deformed $(n+m)$ -permutahedra, not matroid polytope

except 5 cases

THM. There are $\geq 2^{\lfloor \frac{d-1}{2} \rfloor}$ such indecomposable deformed d -permutahedra

$\geq 2^{2^{d-2}}$ (Loho–Padrol–Poullot, arXiv:2510.03177, other method)